DIVISIBILITY OF CONCATENATED ARITHMETIC PROGRESSIONS AND CHAMPERNOWNE PREFIXES

QICHAO WANG* AND DAOYU DONG*

Abstract. We study divisibility questions for integers obtained by concatenation in positional notation. For fixed digit length, concatenating an arithmetic progression is an affine function of its initial term, which yields an exact congruence criterion. We use this to prove a global existence theorem in every even base; in decimal this resolves the existence conjectures recorded in OEIS A075001 and A075000.

We then consider the Champernowne prefixes

$$P(k) = 1 \parallel 2 \parallel \cdots \parallel k.$$

Our main theorem proves that for every positive integer n there exists k such that $n \mid P(k)$, thereby resolving the well-definedness conjecture in OEIS A075002. The proof has two local ingredients. Away from 3, fixed-digit blocks admit unit translations and explicit state periods. At 3, where the unit argument degenerates because $10^d \equiv 1 \pmod 3$, a separate block-shift congruence gives a 3-adic lifting mechanism. The local roots are synchronized at one common prefix index by processing odd prime factors in increasing order; every completed state period has no prime divisor larger than the prime just processed.

1. Introduction

Let

$$P(k) = 1 \parallel 2 \parallel \cdots \parallel k$$

denote the decimal integer obtained by concatenating $1, 2, \ldots, k$, and for $n \geq 1$ let

$$C_n(k) = k \parallel (k+1) \parallel \cdots \parallel (k+n-1).$$

OEIS A075001 asks for the smallest k such that $n \mid C_n(k)$, with value 0 if no such k exists, while OEIS A075000 records the corresponding smallest quotient [4, 3]. OEIS A075002 asks for the smallest k such that $n \mid P(k)$, where P is also the sequence in OEIS A007908 [5, 2]. On September 10, 2026, the three entries still recorded these existence statements as conjectures. We are not aware of a previous proof of the full A075002 assertion; this paper makes no claim of priority.

Our first result is a general even-base construction for concatenated arithmetic progressions. The second and main result is the complete existence theorem for Champernowne prefixes. The proof strategy is modular. We

Date: September 10, 2026.

*These authors contributed equally to this work.

first choose a single sufficiently long digit block. The endpoint is made divisible by the factor of the modulus whose prime divisors divide the base, and the exceptional 3-power is installed by a three-way block shift. The remaining odd prime powers are then installed in increasing order, using state periods to preserve all earlier congruences. The fact that a completed state period has no prime divisor larger than the prime already processed ensures that the next translation remains a unit; a finite capacity bound keeps every chosen endpoint in the same digit block.

Concatenation of arithmetic progressions in arbitrary bases has also been studied by Luca and Teguia Tabuguia, who derive explicit formulae and prove non-holonomicity [1]. Our emphasis is different: divisibility by an externally prescribed modulus and, in particular, the three OEIS existence problems above.

2. Fixed-digit arithmetic progressions

Let $\beta \geq 2$, let $n \geq 1$, let $s \in \mathbb{Z}$, and let $d \geq 1$. Put $B = \beta^d$. Suppose that

$$k,\ k+s,\ \ldots,\ k+(n-1)s$$

are positive d-digit base-β integers. Their concatenation is

$$C_{\beta,n,s,d}(k) = \sum_{j=0}^{n-1} (k+js) B^{\,n-1-j}. \tag{1}$$

Define

$$R_n(B) = \sum_{i=0}^{n-1} B^i, \qquad T_n(B) = \sum_{j=0}^{n-1} j B^{\,n-1-j}.$$

Then

$$C_{\beta,n,s,d}(k) = kR_n(B) + sT_n(B). \tag{2}$$

Proposition 2.1 (Exact fixed-digit criterion)**.** *Let $g = \gcd(R_n(B), n)$. The congruence*

$$n \mid C_{\beta,n,s,d}(k)$$

is solvable in k if and only if

$$g \mid sT_n(B).$$

When solvable, the solutions form one residue class modulo n/g.

Proof. By (2), the desired divisibility is the linear congruence

$$kR_n(B) \equiv -sT_n(B) \pmod{n}.$$

The standard solvability criterion for a linear congruence gives the claim. After division by g, the coefficient $R_n(B)/g$ is invertible modulo n/g. □

Theorem 2.2 (Even-base AP divisibility)**.** *Let $\beta \geq 2$ be even. For every $n \geq 1$ and every $s \in \mathbb{Z}$, there exist $d \geq 1$ and k such that all*

$$k,\ k+s, \ldots, k+(n-1)s$$

are positive d-digit base-β integers and

$$n \mid C_{\beta,n,s,d}(k).$$

Proof. Write

$$n = qm, \tag{3}$$

where q is the largest divisor of n all of whose prime divisors divide β. Thus $(q, m) = 1$ and $(m, \beta) = 1$.

Choose d arbitrarily large subject to

$$q \mid \beta^d, \qquad \beta^d \equiv 1 \pmod{m}.$$

Such d exist: the first condition holds for all sufficiently large d, and the second holds whenever d is a multiple of $\mathrm{ord}_m(\beta)$, with no order condition when $m = 1$.

Put $B = \beta^d$, and let

$$u_- = \min\{0, (n-1)s\}, \qquad u_+ = \max\{0, (n-1)s\}.$$

All blocks have d digits whenever

$$\beta^{d-1} - u_- \leq k \leq \beta^d - 1 - u_+. \tag{4}$$

The length of this interval tends to infinity with d, so we may choose k in it such that

$$k + (n-1)s \equiv 0 \pmod{q}.$$

Modulo q, every term of the sum in (1) except the final block is multiplied by a positive power of B. Hence

$$C_{\beta,n,s,d}(k) \equiv k + (n-1)s \equiv 0 \pmod{q}. \tag{5}$$

Modulo m, since $B \equiv 1$,

$$C_{\beta,n,s,d}(k) \equiv nk + s\frac{n(n-1)}{2} \pmod{m}. \tag{6}$$

The first term is divisible by m. If n is odd, then

$$s\frac{n(n-1)}{2} = mqs\frac{n-1}{2}.$$

If n is even, then $2 \mid \beta$, so the full 2-part of n lies in q; hence $q/2 \in \mathbb{Z}$, and

$$s\frac{n(n-1)}{2} = m\frac{q}{2}\, s(n-1).$$

Thus the concatenation is 0 modulo both coprime factors q and m, hence modulo n. □

Corollary 2.3 (A075001)**.** *For every $n \geq 1$ there exists a positive integer k such that*

$$n \mid k \parallel (k+1) \parallel \cdots \parallel (k+n-1).$$

Hence OEIS A075001 has no zero terms.

Proof. Take $\beta = 10$ and $s = 1$ in Theorem 2.2. □

Corollary 2.4 (A075000)**.** *For every $n \geq 1$, OEIS A075000 is nonzero. More precisely,*

$$A075000(n) = \frac{C_n(A075001(n))}{n}.$$

Proof. Existence follows from Corollary 2.3. It remains only to justify minimality. For fixed n, if $k_2 > k_1$, then comparison of the decimal strings defining $C_n(k_2)$ and $C_n(k_1)$ shows that

$$C_n(k_2) > C_n(k_1),$$

including when one of the indices crosses a power of ten. Thus the least divisible concatenation occurs at the least witnessing index $A075001(n)$. Division by the fixed positive integer n preserves the ordering. □

3. Fixed-block identities for Champernowne prefixes

Fix a digit length d and put $B = 10^d$. Suppose all integers appended after an endpoint k remain in this d-digit block. For $L \geq 0$,

$$P(k+L) = B^L P(k) + \sum_{j=1}^{L} (k+j) B^{L-j}. \tag{7}$$

Set

$$S_L = \sum_{r=0}^{L-1} B^r, \qquad T_L = \sum_{j=1}^{L} j B^{L-j}.$$

Then

$$P(k+L) - P(k) = (B^L - 1)P(k) + kS_L + T_L. \tag{8}$$

A direct telescoping calculation gives

$$(B-1)T_L = BS_L - L. \tag{9}$$

4. Local lifting and state periods

Lemma 4.1 (Unit translation)**.** *Let p be an odd prime with $p \nmid B(B-1)$, let*

$$h = \mathrm{ord}_p(B), \qquad U = T_h,$$

and let $a \geq 0$. If $p \nmid c$ and $L = chp^a$, then

$$P(k+L) - P(k) \equiv cp^a U \pmod{p^{a+1}}, \tag{10}$$

provided all appended integers remain in the same digit block. Moreover $p \nmid U$.

Proof. Modulo p, $B^h = 1$ and $S_h = 0$. From (9),

$$(B-1)U \equiv -h \pmod{p}.$$

Since $h \mid p-1$, both h and $B-1$ are units modulo p, hence so is U.

By the lifting-the-exponent lemma,

$$B^{chp^a} \equiv 1 \pmod{p^{a+1}}.$$

Since $B-1$ is a unit, this also gives

$$S_L \equiv 0 \pmod{p^{a+1}}.$$

Equation (9) therefore yields

$$T_L \equiv -\frac{chp^a}{B-1} \equiv cp^a U \pmod{p^{a+1}}.$$

Substitution into (8) proves (10). □

Lemma 4.2 (State period)**.** *Under the hypotheses of Lemma 4.1, for every $e \geq 1$,*

$$H = p^e \operatorname{ord}_{p^e}(B) \tag{11}$$

is a period of the endpoint state modulo p^e:

$$P(k+H) \equiv P(k) \pmod{p^e}, \qquad k + H \equiv k \pmod{p^e}. \tag{12}$$

Every multiple of H is also such a period. Every prime divisor of H is at most p.

Proof. Let $r = \operatorname{ord}_{p^e}(B)$. Since $r \mid H$, $B^H \equiv 1 \pmod{p^e}$. Because $B-1$ is a unit, $S_H \equiv 0 \pmod{p^e}$. Also $p^e \mid H$, so (9) gives $T_H \equiv 0 \pmod{p^e}$. Equation (8) proves the first congruence in (12), and $p^e \mid H$ proves the second. Iterating the state transition shows that every multiple of H is again a period.

Finally,

$$\operatorname{ord}_{p^e}(B) \mid \varphi(p^e) = p^{e-1}(p-1),$$

so every prime divisor of H is at most p. □

Lemma 4.3 (Exceptional 3-adic block shift)**.** *Let $B = 10^d$, and suppose $v_3(L) = a \geq 1$. Then*

$$P(k+L) - P(k) \equiv Lk + \frac{L(L+1)}{2} \pmod{3^{a+1}} \tag{13}$$

whenever the appended integers remain in one digit block.

Proof. Write $B = 1 + c$. Since $B = 10^d$, we have $9 \mid c$. Expansion of the geometric sum gives

$$S_L - L = \sum_{s \geq 1} c^s \binom{L}{s+1}.$$

Using

$$v_3\binom{L}{m} \geq a - v_3(m)$$

and $v_3(c) \geq 2$, every term on the right has valuation at least $a+1$. Hence

$$S_L \equiv L \pmod{3^{a+1}}.$$

Similarly,

$$T_L - \frac{L(L+1)}{2} = \sum_{s\geq 1} c^s \binom{L+1}{s+2}.$$

From

$$\binom{L+1}{m} = \frac{L(L+1)}{m(m-1)} \binom{L-1}{m-2}$$

we obtain

$$v_3\left(c^s \binom{L+1}{s+2}\right) \geq 2s + a - v_3((s+1)(s+2)).$$

For $s \geq 1$,

$$v_3((s+1)(s+2)) \leq 2s-1;$$

for $s=1$ this is immediate, and for $s \geq 2$ it follows from $3^{2s} > (s+1)(s+2)$. Thus every correction term is divisible by 3^{a+1}.

Finally LTE gives $B^L \equiv 1 \pmod{3^{a+2}}$, so the first term in (8) vanishes modulo 3^{a+1}. Substituting the two congruences for S_L and T_L into (8) proves (13). □

Corollary 4.4 (3-adic lifting)**.** *Let $R = 2^\alpha 5^\beta$, suppose $R \mid k$ and $3^a \mid P(k)$ with $a \geq 1$, and put*

$$L = \operatorname{lcm}(2 \cdot 3^a, R) = 2c3^a, \qquad 3 \nmid c. \tag{14}$$

Then exactly one of

$$k, \qquad k+L, \qquad k+2L$$

is a root modulo 3^{a+1}, while all three endpoints remain divisible by R. Moreover

$$2 \cdot 3^e \tag{15}$$

is a state period modulo 3^e.

Proof. The preceding lemma gives

$$\frac{P(k+L) - P(k)}{3^a} \equiv c(2k+1) \pmod 3.$$

Since $3 \mid P(k)$ and $10 \equiv 1 \pmod 3$,

$$0 \equiv P(k) \equiv 1 + 2 + \cdots + k = \frac{k(k+1)}{2} \pmod 3,$$

so $k \equiv 0$ or $2 \pmod 3$. Hence $2k+1 \not\equiv 0 \pmod 3$, and the three choices give the three possible residues at the next 3-adic level. Because $R \mid L$, the endpoint condition modulo R is preserved.

Taking $L = 2 \cdot 3^e$ in Lemma 4.3 gives $P(k+L) \equiv P(k) \pmod{3^e}$ and $k + L \equiv k \pmod{3^e}$, proving the period assertion. □

5. A common prefix for every modulus

Theorem 5.1 (A075002)**.** *For every positive integer n, there exists $k \geq 1$ such that*

$$n \mid P(k).$$

Consequently OEIS A075002 is well-defined for every n.

Proof. Factor

$$n = R\,3^e \prod_{i=1}^{r} p_i^{e_i}, \qquad R = 2^\alpha 5^\beta, \qquad 7 \leq p_1 < \cdots < p_r. \tag{16}$$

Let

$$D = \operatorname{lcm}_{1 \leq i \leq r} \operatorname{ord}_{p_i^{e_i}}(10),$$

with $D = 1$ if $r = 0$. We shall work in one sufficiently long d-digit block, where

$$d \equiv 1 \pmod{D}, \qquad d \geq \max(\alpha, \beta). \tag{17}$$

Such d are arbitrarily large. Put $B = 10^d$. Then for every i,

$$B \equiv 10 \pmod{p_i^{e_i}}.$$

We first make the capacity choice precise. For fixed n, every step length appearing below is obtained by taking least common multiples of R, prime powers dividing n, and multiplicative orders modulo those prime powers. Hence all possible step lengths, and the maximum total displacement obtained by making at most $p-1$ choices at each p-adic stage, are bounded by a finite constant $W(n)$ independent of d. Let

$$M_0 = R \cdot 3^{\min(e,1)}. \tag{18}$$

Choose d in the congruence class in (17) so large that

$$9 \cdot 10^{d-1} > M_0 + W(n). \tag{19}$$

Let k_0 be the least multiple of M_0 that is at least 10^{d-1}. Then $0 \leq k_0 - 10^{d-1} < M_0$, and (19) keeps the entire construction below inside this single d-digit block.

Since $R \mid k_0$ and $B \equiv 0 \pmod{R}$,

$$P(k_0) = BP(k_0 - 1) + k_0 \equiv 0 \pmod{R}. \tag{20}$$

If $e > 0$, then $3 \mid k_0$, and because $10 \equiv 1 \pmod{3}$,

$$P(k_0) \equiv \frac{k_0(k_0+1)}{2} \equiv 0 \pmod{3}.$$

If $e > 1$, repeatedly apply Corollary 4.4. At stage a, use

$$L_a = \operatorname{lcm}(2 \cdot 3^a, R).$$

One of the three choices raises a root modulo 3^a to a root modulo 3^{a+1}, and the R-condition is preserved. Thus we obtain an endpoint k with

$$R3^e \mid P(k).$$

If $e > 0$, set $H = 2 \cdot 3^e$; otherwise set $H = 1$. In either case H is a common state period for all already installed odd prime-power conditions.

Now process $p_1, \dots, p_r$ in increasing order. Suppose all preceding prime powers have been installed at the current endpoint k, and H is a common state period for them. Let $p = p_i$ and

$$h = \operatorname{ord}_p(10).$$

Since $B \equiv 10 \pmod{p}$, we also have $h = \operatorname{ord}_p(B)$, and $p \nmid B(B-1)$.

Set

$$J_0 = \operatorname{lcm}(h, H, R). \tag{21}$$

Every prime factor of the period contributed by an earlier prime $q < p$ is at most q, while R has no prime divisors other than 2 and 5. Therefore $p \nmid HR$, and also $p \nmid h$ since $h \mid p-1$. Hence $J_0 = c_0 h$ with $p \nmid c_0$. Lemma 4.1 with $a = 0$ shows that

$$P(k + tJ_0), \qquad 0 \le t < p,$$

runs through all residues modulo p. Because J_0 is a multiple of H and R, every previously installed congruence and the endpoint condition modulo R are preserved. Choose the unique t giving a root modulo p.

Suppose inductively that a root modulo p^a has been obtained, where $1 \le a < e_i$. Set

$$J_a = \operatorname{lcm}(hp^a, H, R). \tag{22}$$

Again $p \nmid HRh$, so $J_a = c_a h p^a$ with $p \nmid c_a$. Lemma 4.1 gives

$$P(k + J_a) - P(k) \equiv c_a p^a U \pmod{p^{a+1}}, \qquad p \nmid U.$$

Thus exactly one of

$$k,\ k + J_a, \dots, k + (p-1)J_a$$

is a root modulo p^{a+1}, while all earlier congruences remain intact. Repeating installs p^{e_i}.

After completing this prime power, replace

$$H \leftarrow \operatorname{lcm}(H, p^{e_i} \operatorname{ord}_{p^{e_i}}(B)). \tag{23}$$

By Lemma 4.2, the new factor is a state period modulo p^{e_i}. Since $B \equiv 10 \pmod{p_i^{e_i}}$, it equals $p^{e_i} \operatorname{ord}_{p^{e_i}}(10)$. Every prime divisor of this factor is at most p, so the same argument applies at the next, larger prime.

At termination, the same endpoint k satisfies every prime-power condition in the factorization (16). Therefore $n \mid P(k)$. □

Remark 5.2. The increasing-prime order is essential to this proof. A single global affine orbit need not have unit translation simultaneously at all prime factors. The construction above avoids that obstruction by preserving completed states with periods whose prime divisors are all at most the prime just processed; a larger, unprocessed prime therefore cannot divide the next step's coefficient.

6. Verification and provenance

The proofs are independent of computation. A separate verifier evaluates prefixes modulo the requested modulus using fixed-block transition matrices and cross-checks each matrix jump against a literal recurrence on small instances. The recorded checks include the published initial values of A075002, the A075001 construction for $n \leq 2000$, both lifting lemmas, and common-index constructions for $n \leq 500$, together with the five composite moduli below 5000 for which a single affine orbit cannot provide unit translations at all prime factors. These computations are regression and reproducibility checks, not premises of the proofs.

The source, bibliography, verifier, verification record, and independent audit are preserved in the public repository `oeis-papers`. The frozen release `a075000-a075002-v1.0.1` contains this paper as a stable snapshot. From the paper directory, the verifier can be reproduced with

```
python verify_concatenation_divisibility.py
```

The manuscript grew out of a larger computer-assisted investigation, but that research history is not used as a premise here. The OEIS status statement above records only the checked date; it is not intended as a claim of literature-wide priority.

References

1. F. Luca and B. Teguia Tabuguia, *Concatenations of terms of an arithmetic progression*, Journal of Difference Equations and Applications **32** (2026), no. 4, 437–456, DOI: 10.1080/10236198.2025.2603385; preprint: arXiv:2201.07127.
2. OEIS Foundation Inc., *A007908: Concatenation of the positive integers*, The On-Line Encyclopedia of Integer Sequences, OEIS entry A007908, accessed September 10, 2026.
3. ———, *A075000: Smallest number such that $na(n)$ is a concatenation of n consecutive integers*, The On-Line Encyclopedia of Integer Sequences, OEIS entry A075000, accessed September 10, 2026.
4. ———, *A075001: Smallest k such that the concatenation of n consecutive numbers starting with k is a multiple of n*, The On-Line Encyclopedia of Integer Sequences, OEIS entry A075001, accessed September 10, 2026.
5. ———, *A075002: Smallest k such that* A007908(k) *is divisible by n*, The On-Line Encyclopedia of Integer Sequences, OEIS entry A075002, accessed September 10, 2026.